6 three digit numbers are 123, 132, 231, 213, 312, 321.

Notations:If $n \geq 1$ and $0 \leq r \leq n$, then the number of all permutations of n distinct things taken r at a time is denoted by

$$P(n,r) \text{ (or) } {}_nP_r$$

We have the following theorem *without proof.*

Theorem: $np_r = \dfrac{n!}{(n-r)!}$

Example 2.15

Evaluate: $5P_3$ and $P(8, 5)$

Solution

$$5P_3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = 60.$$

$$P(8, 5) = 8P_5 = \frac{8!}{(8-5)!} = \frac{8!}{3!} = \frac{8\times7\times6\times5\times4\times3!}{3!} = 6720$$

Results

(i) $0! = 1$

(ii) $np_0 = \dfrac{n!}{(n-0)!} = \dfrac{n!}{n!} = 1$

(iii) $np_1 = \dfrac{n!}{(n-1)!} = \dfrac{n(n-1)!}{(n-1)!} = n$

(iv) $np_n = \dfrac{n!}{(n-n)!} = \dfrac{n!}{0!} = n!$

(v) $np_r = n(n-1)\,(n-2)\,\ldots\,[n-(r-1)]$

Example 2.16

Evaluate: (i) $8P_3$ (ii) $5P_4$

Solution

(i) $8P_3 = 8 \times 7 \times 6 = 336.$

(ii) $5P_4 = 5 \times 4 \times 3 \times 2 = 120.$

Example 2.17

In how many ways 7 pictures can be hung from 5 picture nails on a wall ?

Solution

The number of ways in which 7 pictures can be hung from 5 picture nails on a wall is nothing but the number of permutation of 7 things taken 5 at a time

$$= 7P_5 = \frac{7!}{(7-5)!} = 2520 \text{ ways.}$$

Example 2.18

Find how many four letter words can be formed from the letters of the word "LOGARITHMS" (words are with or without meanings)

Solution

There are 10 letters in the word LOGARITHMS

Therefore $n = 10$

Since we have to find four letter words,

$$r = 4$$

Hence the required number of four letter words

$$= np_r = 10\,p_4$$

$$= 10 \times 9 \times 8 \times 7$$

$$= 5040.$$

Example 2.19

If ${}_nP_r = 360$, find n and r.

Solution

$${}_nP_r = 360 = 36 \times 10$$

$$= 3 \times 3 \times 4 \times 5 \times 2$$

$$= 6 \times 5 \times 4 \times 3 = 6P_4$$

Therefore $n = 6$ and $r = 4$

02_11th_BM-STAT_Ch-2-EM.indd 33 21-04-2020 12:18:23 PM